

Davi Guanabara Aragão

AI/ML Engineer and Ph.D. Researcher in Autonomous Systems

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Lockheed Martin —Artificial Intelligence Center

Dear Hiring Manager,

Over the past five years, I have intentionally built my academic and research path around a single goal: developing artificial intelligence for autonomous aerospace systems.

This path began when I joined the Aeronautics Institute of Technology (ITA) to pursue an M.S. in Aeronautical Computing. During my master's research, I joined the CTEDS (Cooperative Threat Engagement with Heterogeneous Drone Swarms) research program, an international Brazil-Sweden collaboration across government, academia, and industry. Working with researchers from ITA and Embraer, I studied autonomous decision-making for cooperative Loyal Wingmen using deep reinforcement learning and graph neural networks.

Through this work, I studied the state of the art in autonomous aerial combat, including DARPA's Air Combat Evolution (ACE) program, the AlphaDogfight Trials, and recent advances in reinforcement learning for autonomous combat systems. That research led to the Best Brazilian Military Master's Dissertation Award and two peer-reviewed publications, including one in the Journal of Intelligent & Robotic Systems (Springer Nature).

To broaden my expertise beyond reinforcement learning, I completed a Graduate Certificate in Data Science at ITA, where I built machine learning systems for geospatial environmental prediction with an emphasis on reproducible experimentation and software engineering. I am now pursuing a Ph.D. in Electronic Engineering and Computing at ITA, collaborating with the Brazilian Airspace Control System (SAC/ICEA) on research in urban airspace management, geospatial optimization, and intelligent software for Brazil's future UTM ecosystem.

Across these projects, I have deliberately expanded my work from autonomous air combat to geospatial AI, intelligent transportation systems, and software platforms for operational decision-making. Although each project addresses a different application, they share the same objective: designing reliable intelligent systems that can support complex aerospace operations under uncertainty.

What draws me to the Lockheed Martin Artificial Intelligence Center is the opportunity to contribute to the next generation of autonomous aerospace systems. Throughout my academic path, I have consistently pursued the same class of problems that organizations like Lockheed Martin address: autonomous decision-making, intelligent mission systems, and artificial intelligence for complex operational environments. I believe my background in reinforcement learning, multi-agent systems, and aerospace research aligns naturally with LAIC's mission, and I would be excited to continue growing as an engineer while contributing to challenging real-world defense applications.

Thank you for your time and consideration. I look forward to the opportunity to discuss how my background and research experience could contribute to the work being done at Lockheed Martin.

Sincerely,

Davi Guanabara Aragão